Mohammed Ragab Al-Attar

Mechanical Engineer

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Summary

An analytical Mechanical Engineer with specialized expertise in **energy systems**, **heat transfer**, and **mechanical design**, backed by a Master's degree in Biomass Conversion. Proficient in leading diverse engineering projects, from MEP and aerospace thermal design to conducting **condition monitoring**, **reliability studies**, and **root cause failure analysis** for rotating equipment in the refining sector. Leverages strong programming skills in **Python**, **MATLAB**, and LabVIEW for simulation, control systems, and data analysis to **enhance system performance and reliability**. Seeking a challenging role that utilizes this diverse technical skill set to develop innovative engineering solutions.

Education

Master of Science in Biomass Conversion Energy & Measurements Control Engineering | 2023 – 2025
Ain Shams University, Cairo
GPA: 3.53 |Research Title: “Development of a decentralized agricultural waste to solid fuel conversion system”
Bachelor of Engineering in Power Mechanical | 2017 – 2022
Ain Shams University, Cairo
Cumulative Grade: Very Good with Honor (Top Ranked)
GPA: 3.3 | Graduation Project Grade: Excellent

Professional Experience

Reliability & Rotating Equipment Engineer | Egyptian Refining Company | 2023 – Present
- Implement and manage condition monitoring programs for critical rotating equipment (pumps, compressors, turbines) using vibration analysis, oil analysis, and thermal imaging to preemptively detect faults.
- Lead Root Cause Analysis (RCA) investigations for equipment failures, identifying primary failure modes and developing corrective action plans to prevent recurrence.
- Develop and optimize maintenance strategies (preventive, predictive, and corrective) to improve equipment reliability, extend asset lifespan, and reduce operational downtime.

Structural & Thermal Engineer (Part-Time) | Advanced Rocket Technologies (ART) | 2023 – 2024
- Performed Finite Element Analysis (FEA) on structural components of low-cost launch systems to validate design integrity under mechanical and thermal loads.
- Conducted heat transfer simulations to analyze the thermal performance of critical systems, ensuring they operated within safe temperature limits during launch and flight.

Maintenance Engineer | Cairo Metro – Third Line | 2022
- Executed the preventive maintenance program for large-scale HVAC systems across multiple metro stations, servicing chillers, Air Handling Units (AHUs), and Fan Coil Units (FCUs).

Academic & Research Projects

Team Leader – MEP Design Project (Graduation Project) | 2021 – 2022
- Led a team in the end-to-end design of a balanced MEP system (HVAC, Firefighting, Water Supply, Drainage) for a multi-building hospital complex.
- Utilized Autodesk Revit for 3D modeling and system integration, and AutoCAD and HAP for detailed design, load calculations, and duct sizing.

Team Leader – Power Plant Design Project | 2021 – 2022
- Directed the design and selection of a sustainable energy and water solution for a new 200,000 Feddan desert reclamation community in the El-Moghra region.
- Analyzed and proposed an optimal mix of energy generation sources to meet the community's power demands reliably and sustainably.

Research Associate – CoAgrow+ Biogas Project | 2022-2023
- Designed and constructed a lab-scale anaerobic digester to investigate and optimize the co-fermentation process of Egyptian agricultural waste.

Teaching Assistant – Torrefaction Project | 2022 - 2023
- Assisted in the development of a biochar system designed for the optimal torrefaction and energy conversion of local agricultural waste.

Professional Training

Trainee, Satellite Systems | Egyptian Space Agency | 2021

- Selected for an intensive 11-week government training program covering all theoretical and practical aspects of space technology.
- **Achieved First Place in the Mechanical Department** for a hands-on project focused on the design and practical application of a CubeSat deployment mechanism and antenna pointing system.

Skills

- **Technical Expertise:**
  - Condition Monitoring & Reliability Analysis
  - Root Cause Analysis (RCA)
  - Mechanical Design & Prototyping
  - Heat Transfer & Thermodynamics
  - MEP Systems Design (HVAC, Firefighting, Plumbing)
  - Energy Systems Analysis & Optimization
- **Software & Tools:**
  - **CAD/CAE:** SolidWorks, AutoCAD, Autodesk Revit MEP, ANSYS, Inventor Pro
  - **Energy & Process:** Homer Pro, ASPEN Plus, Energy Plan, WANDA Pipeline, Duct Sizer
  - **Project Management:** Primavera P6, MS Excel (including VBA), MS Word
  - **Creative:** Adobe Photoshop, Illustrator
- **Programming & Data:**
  - **Languages:** Python, MATLAB, C++/Java, Octave
  - **Specialties:** AI/Machine Learning Development, Data Analysis, PLC/Ladder Logic, LabVIEW

Professional Development & Certifications

- **Industry Training & Internships:**
  - **Enppi Academy Training** (3 Months, 2022): Intensive program covering Process Engineering, Piping, Static & Rotating Equipment, Instrumentation, Inspection, and Project Management.
  - **Supply Chain Internship** | Hassan Allam (2021): Gained practical experience in Forecasting, Inventory Management, Linear Programming, and Warehouse Management.
- **Harvard University (via edX):**
  - **CS50's Introduction to Artificial Intelligence with Python (CS50-AI)** (2020)
  - **CS50's Introduction to Computer Science (CS50x)** (2019)
  - **CS50 for Technology Professionals (CS50-TECH)** (2021)
- **Technical & Professional Certifications:**
  - **MEP Revit Course** | Engineers' House (2022): Comprehensive training in HVAC, Firefighting, and Water Supply system design using Revit.
  - **Project Management Path** | LinkedIn Learning (2020): Completed a 12-hour learning path covering core project management principles.
  - **Energy Plan & Homer Pro** | Online Course (2022): Focused on Load Estimation, Generation Sizing, and resource optimization.
  - **MATLAB for Engineers** | Udemy (2021): Covered advanced use of variables, matrices, flow control, and solving mathematical equations.
- **Language Proficiency:**
  - **IELTS Academic Test** | British Council (2018): Achieved a 6.5 Overall Band Score.

Leadership Experience & Activities

Business Plan Competition (BLC) Leader | ASU Racing Team | 2022

- Led the business team in developing a comprehensive startup plan for an automotive product, covering market analysis, financial projections, and production strategy.
- Directed the creation and delivery of the final business plan presentation for competition judges.

Founder & CEO | Semi-Colon Organization | 2020 – 2022

- Founded a student-led organization to bridge the gap between academic and practical industry skills.
- Organized workshops and training programs to prepare students for technical and non-technical roles in local and international competitions.

Languages

- **English:** C1 (Excellent) **French:** B1 (Good) **Arabic:** Native